

Executive Summary

Research in cognitive psychology strongly guides how to structure your DCS professional development (PD). Evidence shows that **multiple short, spaced sessions** far outperform one long cram session ¹ ² . In practice, this means breaking your Thursday/Friday study blocks into focused chunks on different topics (video learning, flashcards/quizzes, scenario practice, role-play, content creation, etc.), rather than trying to “power through” single tasks for hours. Similarly, **frequent active recall** (quiz questions, flashcards, or explaining concepts aloud) dramatically strengthens memory ³ ⁴ . Mixing up activities (interleaving video, hands-on tasks, and writing) helps too ⁵ .

At the same time, real-world interruptions (quick tickets, walk-ups, ad-hoc tasks) impose a **cognitive switching cost** ⁶ . Research finds that even a brief interruption (“context switch”) adds delay, especially for complex tasks. To “degrade gracefully,” schedule work in independent 30–60 minute blocks that can stand alone. If a block is cut short, the plan simply shifts remaining tasks into the next available slot, rather than derailing the entire schedule. Build buffers for recovery (10–15 minute breathing breaks) to reset focus after a tough switch.

Based on these principles, the revised prompt for Claude emphasizes: (1) your actual CompTIA progress and remaining hours, (2) a broad definition of PD time (videos, DCSPrep modules, app-building, ticket templates, etc.), (3) the goal of improving DCS-level 1 support and soft skills, and (4) realistic interruptions and flexibility. It also asks Claude to research evidence-based scheduling strategies (spacing, retrieval, interleaving) and how to incorporate them into your 13+ hours/week schedule.

Key deliverables from Claude should include a prioritized list of research questions, schedule templates (light/medium/heavy) that flex with interruptions, specific app enhancements (e.g. next-action recommendations, rubric grading), and measurable success criteria (e.g. PD hours, readiness scores, ticket-quality metrics).

Revised Prompt for Claude

Below is a ready-to-run prompt for Claude, incorporating all the needed details:

Act as a senior full-stack Next.js/TypeScript engineer, learning-science specialist, and professional development coach.

You are optimizing DCSPrep – a local-first PD app – for a DCS Level-1/aspiring Level-2 support technician (Josh at Dubbo Christian School). The user has already completed all CompTIA A+ Core 1 videos (10h11m) and is currently in Core 2 (up to section 1.6 on Windows Settings), with roughly 10 hours of Core 2 content remaining.

Josh has approximately 13+ hours of formal PD time each week (Thursday and

Friday blocks), which can be used flexibly any time he is **not** handling live support tickets or tasks assigned by his supervisor. "Professional development" includes anything tech- or support-related: watching training videos, doing DCSPrep learning modules, quizzes, flashcards, scenario labs, AI roleplays, writing ticket notes or support checklists, building DCSPrep content, and recording evidence in a PD log.

Important constraints: interruptions happen frequently (5-minute walk-ups, 20-minute tickets, or even an hour-long urgent task). The schedule must **degrade gracefully** - partially completed study tasks should resume later. ADHD and fatigue mean blocks should include short breaks.

Your mission is to research and recommend an **evidence-based weekly PD plan** and app features to maximize Josh's learning and performance. Specifically:

- ****Learning science:**** How should spacing, retrieval practice, interleaving, and other cognitive principles shape the schedule? (Cite primary/academic sources.)
- ****PD schedule:**** What is the optimal mix of activities (video learning, practice quizzes/flashcards, scenario labs, roleplay, content creation, review) over ~13 hours per week? Provide at least three templates (light, balanced, intensive) with time allocations. Show how each adapts if blocks are interrupted.
- ****Flexible learning:**** How to build resilience to interruptions (breakdown tasks, use partial blocks, quick "micro-learning" activities)?
- ****App coaching logic:**** How should the DCSPrep app suggest "next best actions" and evidence-gathering workflows? Recommend new features: e.g. a study recommendation engine (based on weak topics, due items, readiness), automated ticket-note rubric grading, "repair packs" for weak areas, troubleshooting playbooks and asset profiles, improved evidence export. Specify file/route names or types if possible (e.g. `src/lib/studyPath.ts`, new pages `/playbooks` and `/assets`, rubric grader module, etc.).
- ****PD goal:**** Emphasize DCS-specific goals (improving Level-1 support skills, ticket quality, escalation decisions, progress toward A+/M365 mastery), rather than general self-help.
- ****Measurable outcomes:**** What metrics will indicate success? (e.g. % of planned PD time used, completion of modules, readiness score, ticket-note quality improvements, scenario completion count, etc.)

Return a structured report with these sections:

- Executive Summary
- The Prompt (as above, ready to use)
- Ranked Research Questions (at least 12, prioritized)
- Weekly Schedule Templates (light, balanced, intensive; explain how interruptions are handled)
- Recommended App Logic/Feature Changes (with file or route suggestions)
- 4-Week Study Cycle Timeline (mermaid diagram)
- Activity Mix Comparison Table (video vs practice vs roleplay vs building vs

review, minutes/week)

- Measurable Success Criteria (at least 8, quantifiable)

Prioritized Research Questions for Claude

1. **Effective Scheduling:** How can Josh structure his 13+ weekly PD hours (broken into Thu/Fri blocks) to use spacing and retrieval practice optimally? (E.g. dividing tasks into daily 60–90 min sessions vs. longer blocks.)
2. **Task Mix:** What balance of learning activities (video lectures, hands-on labs, flashcards/quizzes, scenario practice, role-play, content creation, reflection) yields the best learning for technical support skills ² ⁴ ?
3. **Spacing & Retrieval:** How often should flashcards or quizzes be repeated (spaced intervals) to maximize retention of CompTIA/IT knowledge? (Use evidence on scheduling review intervals ⁷ ³ .)
4. **Interleaving Content:** How should different topics be interleaved across and within study blocks? (For example, mixing Core 2 topics, network basics, and DCSPrep tasks to enhance transfer ⁵ .)
5. **Interruptions:** What strategies help “degrade gracefully” under interruptions? (E.g. plan short milestones, use quick micro-tasks, schedule buffer time; reduce cognitive switching costs ⁶ .)
6. **Active Recall:** How to incorporate retrieval practice seamlessly? (E.g. end each block with a self-quiz or flashcard session ³ ⁴ .)
7. **ADHD & Fatigue:** What work–break cycle or environment adjustments help sustain focus? (Research on short breaks, exercise, routine to combat cognitive fatigue.)
8. **App “Next Best Action”:** How to design DCSPrep’s recommendation engine? (What data to use: progress, weak topics, due items, calendar; how to rank tasks by priority.)
9. **Rubric Grading:** What criteria and feedback elements should the app use to grade scenario answers and ticket notes? (E.g. clarity of symptom, scope, evidence, safe action, escalation step.)
10. **Repair Packs & Playbooks:** Which real-world DCS issues (printers, login, network, etc.) should have custom playbooks? (How to structure them for quick lookup.)
11. **Evidence Tracking:** What evidence (hours logged, quizzes passed, modules done) best reflects learning progress? (How to measure “readiness” for Level 2.)
12. **Motivation & Self-Driven PD:** What adult-learning principles (e.g. self-directed learning, intrinsic motivation) fit an ICT professional’s schedule? (How to keep PD engaging and relevant to daily work.)

Weekly PD Schedule Templates

Each template assumes Josh has ~13 hours/week (780 minutes) on Thu–Fri, with tasks in 30–90 minute blocks. Blocks can be broken into “half-blocks” if interrupted. Light, Balanced, and Intensive variants are shown; all embed spacing (multiple shorter sessions) and retrieval practice. Each also reserves short breaks (e.g. 5–10 min between tasks) to reset focus.

- **Light Schedule (~10 h/week):** ~600 min. (e.g. final-year university student pace)
- **Thursday:** 2h (120 min) – Watch Core 2 videos (sections 1.6–1.7), take notes; 1h (60 min) – DCSPrep Module (e.g. Windows settings); 30 min – **Flashcards/Quiz** on last week’s topics; 30 min break/Q&A.
- **Friday:** 3h (180 min) – Continue Core 2 videos (1.8–2.0); 1h (60 min) – Scenario Lab (e.g. “Wi-Fi issue”); 30 min – Roleplay (AI coach on printer issue); 30 min – **Flashcards** review; 30 min – Reflection/PD log.

- *Interruptions:* If a 20-min ticket arrives during a video, finish the remaining video later. If a 5-min walk-up arrives, pause current task and resume. Critical: each 30–60 min block can stop mid-task; leave a small “checkpoint” note or video timestamp to resume smoothly.

- **Balanced Schedule (~13 h/week):** ~780 min. (expected PD pace)

- *Thursday:* 2h (120 min) – Core 2 videos (2.0–2.1) + note-taking; 1h (60 min) – **Quiz/Flashcards** (retrieve yesterday’s material); 1.5h (90 min) – DCSPrep module (Networking basics) + summary note; 30 min – Break/Review log; 1h (60 min) – Scenario Lab (e.g. “Mapped drive missing”).
- *Friday:* 2h (120 min) – Core 2 videos (2.2–2.3); 1h (60 min) – **Flashcards** on Core 2 topics; 1h (60 min) – Roleplay (teacher with login issue); 1h (60 min) – Practical output (create ticket template for printer issue); 30 min – DCSPrep quiz/knowledge check.
- *Interruptions:* Keep all tasks ≤ 1.5 h. If a half-block (e.g. 1h) is interrupted by a support call, use remaining time next day. Maintain a flexible “to-do” list. For example, if Scenario Lab is cut short, continue on Friday instead of new video.

- **Intensive Schedule (~16 h/week):** ~960 min. (for accelerated progression)

- *Thursday:* 3h (180 min) – Core 2 videos (2.3–2.5) + detailed notes; 1h (60 min) – flashcards + mini-quiz (active recall); 1h (60 min) – DCSPrep module (Group Policy) + write escalation note; 30 min – Break + quick review; 1h (60 min) – Scenario Lab; 30 min – Review log and plan.
- *Friday:* 3h (180 min) – Core 2 videos (2.6–3.0); 1.5h (90 min) – **Interleaved practice:** alternate 30 min segments of flashcards, roleplay, and coding/practical building; 1h (60 min) – Create a Troubleshooting Playbook (e.g. Wi-Fi issue); 30 min – Break; 1h (60 min) – DCSPrep module (review weak areas); 30 min – PD log/journal and rest.
- *Interruptions:* Similar flex blocks. If an entire block is lost (e.g. an urgent issue takes 3 hours), push remaining tasks to next week’s plan or shorten next break to compensate. For example, if half of Thursday’s video time is consumed, finish it Friday morning before starting new topics.

All templates assume recovery after interruptions: e.g., a 5–10 min break after a ticket, then resume a related or different task. Tasks are mixed (videos vs. hands-on vs. writing) to leverage interleaving ⁵. Each template also schedules low-stakes recall (flashcards/quizzes) at the end of each day to consolidate learning

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```

timeline
    title 4-Week PD Cycle (Illustrative)
    Week 1 : Core2 Videos (sec 1.6–2.1) : DCSPrep Modules (Security) : Flashcard reviews
    Week 2 : Core2 Videos (sec 2.2–2.4) : Scenario Labs (network, printer) : Roleplays (teacher/sound)
    Week 3 : Core2 Videos (sec 2.5–3.0) : Troubleshooting Playbooks creation : Building ticket templates
    Week 4 : Review Core2 + CompTIA quizzes : Advanced Scenario Labs : Evidence Pack & Reflection

```

Activity Mix Comparison

Activity	Light (min/week)	Balanced	Intensive
Video Learning (Prof. Messer videos, etc.)	180 (3h)	240 (4h)	300 (5h)
Retrieval Practice (Flashcards/Quizzes)	90 (1.5h)	120 (2h)	150 (2.5h)
Scenario Labs / Roleplay (Interactive practice)	120 (2h)	210 (3.5h)	270 (4.5h)
Practical Outputs (Templates, scripts, modules)	90 (1.5h)	150 (2.5h)	180 (3h)
Review/Reflection (Notes, PD log)	60 (1h)	60 (1h)	60 (1h)
Total	600 (10h)	780 (13h)	960 (16h)

In all cases, time is split across **Thursday and Friday** blocks. The Balanced plan (~780 min) is roughly 13 h, matching your available PD time. Each category follows best practices: e.g. flashcards occur every day (retrieval practice ⁴), scenarios and role-plays are limited to allow focus, and video sessions are broken into ≤ 90 -min segments (to avoid fatigue). The Intensive plan (~16 h) pushes deeper engagement, while the Light plan (~10 h) is a manageable minimum.

Recommended App Logic and Features

To support this PD schedule and make learning “sticky,” DCSPrep should implement the following **app logic changes** and new features:

- **Study Recommendation Engine** (`studyPath.ts`): Create `src/lib/studyPath.ts` to analyze Josh’s progress and weak areas, then suggest prioritized tasks (“next best action”). Consider inputs like quiz scores, topic weights, due flashcards, and scenario performance. Expose this on Dashboard/Focus as a “Today’s Top Recommendations” card.
- **Rubric-based Grading** (`rubricGrader.ts`): Add a grading module (e.g. `src/lib/rubricGrader.ts`, types in `src/types/grading.ts`) to score written answers and ticket notes against a rubric. Criteria: clarity of symptom, scope, evidence, safety, escalation. Integrate this into Scenario Lab final steps and short-answer quizzes. Display the score, strengths/gaps, and an improved example answer after submission.
- **Retrieval Scheduling** (`spacedRepetition.ts`): Leverage or extend existing spaced repetition code to schedule flashcard reviews at optimal intervals (embedding retrieval practice into the schedule). Ensure flashcards from older modules reappear in the PD plan when due.
- **Practice Quizzes**: Ensure short-answer and scenario-response questions use the new rubric for feedback and that retries or guided hints are available for missed items.
- **Troubleshooting Playbooks**: Introduce a new custom data type (uploadable JSON) for common school IT issues. Example file:

```
type TroubleshootingPlaybook = {
  id: string; title: string; domain: string; level: string; symptoms:
  string[]; firstQuestions: string[]; safeChecks: {id,title,steps}[];
```

```
escalationTriggers: string[]; doNotDo: string[]; ticketTemplate: string;
}
```

Add upload support via Settings and pages at `/playbooks` and `/playbooks/[id]`. Populate starter playbooks (e.g. “Wi-Fi but no Internet”, “Printer not releasing”, “User login error”).

- **Asset/System Profiles:** Create a new custom type for “asset profiles” (printer models, ViewBoard setup, software platforms). For example:

```
type DcsAssetProfile = {
  id: string; name: string; category: "device"|"printer"|"network"|...;
  commonSymptoms: string[]; safeChecks: string[]; usefulTools: string[];
  escalationOwner?: string; level1Boundaries: string[]; privacyNotes:
  string[];
}
```

Allow uploads and browsing at `/assets` and `/assets/[id]`. These provide context (e.g. “Staff Laptop”, “Student iPad Trolley”, “MFD printer network share”) to guide safe troubleshooting steps.

- **Enhanced Evidence Pack:** Upgrade the Evidence Pack page (`/evidence-pack`) to export a **PDF/print-friendly report** summarizing PD progress. Include customizable date ranges and sections for modules completed, quiz scores, scenarios solved, cards reviewed, and teacher notes. Remove any student-sensitive details and format for management review.
- **Due-Today & Focus pages:** Modify `/due-today` and `/focus` to show spaced-review items (flashcards/quizzes due) and to highlight the next study recommendation. For example, list “Review TCP/IP flashcards (due)” alongside “Complete DCSPrep networking module”.
- **Academic PD Integration:** Link the user’s CompTIA A+ plan (Core2) to DCSPrep’s Academic PD section. For instance, tag relevant modules (network, endpoints) so that progress in CompTIA is tracked in Readiness.
- **Customization & UX:** Allow toggling of the new Coaching features (rubric on/off) and show concise tips (e.g. “Remember to space your reviews!”). Ensure mobile-friendly layout, clear interruption-resume cues (like timestamps on tasks), and avoid any language that isn’t relevant to DCS context.

Each change should preserve existing data. For uploads, expand `src/lib/customModules.ts` to recognize the new types (playbooks, profiles). Create new database-like JSON collections (e.g. `playbooks.json`, `assets.json`) and merge with defaults. Add needed routes/components under `app/` (e.g. `app/playbooks/page.tsx`, `app/playbooks/[id]/page.tsx`, `app/assets/page.tsx`, `app/assets/[id]/page.tsx`), and update navigation/search indexing accordingly.

Measurable Success Criteria

1. **PD Time Utilization:** Achieve $\geq 90\%$ of scheduled PD block hours each week (e.g. log $\geq 12/13$ h per week in PD activities).
2. **CompTIA Progress:** Complete all remaining Core 2 content within 4 weeks (10 hours of video), on schedule.

3. **Module Completion:** Finish each planned DCSPrep module (e.g. Windows Control Panel, Group Policy) by its target week.
4. **Scenario Practice:** Complete at least one new Scenario Lab every week. Aim for all ~10 new scenario prompts done in a month.
5. **Readiness Score:** Increase overall readiness rating (as computed by the app) by at least 20 points within 4 weeks, especially in weak topics (networking, security, etc.).
6. **Ticket-Note Quality:** Average rubric-based ticket note score improves to “competent/strong” (e.g. >80%) on post-scenario writing exercises (test at start vs. end of 4 weeks).
7. **Flashcard Retention:** Maintain $\geq 80\%$ correct recall on due flashcards across weeks (showing effective spaced review).
8. **Practical Outputs:** Produce at least 4 PD artifacts (e.g. playbook, checklist, cheat sheet, ticket template) by week 4 and incorporate them into daily support.

Meeting these criteria will demonstrate that the revised schedule and app logic are working: Josh will have used his PD time effectively, retained critical knowledge, and applied it to improve his support skills and documentation.

Sources: Cognitive science research underpins these recommendations ¹ ³ ⁵ ⁶ . The strategy leverages spacing, retrieval practice, interleaving, and minimizes task-switching costs to maximize learning.

¹ ⁷ The most common question I get asked about retrieval practice (and the answer)

<https://www.retrievalpractice.org/strategies/optimal-spacing>

² ³ Retrieval and Spaced Practice: Study Strategies That Must Be Combined - Evidence Based Education

<https://evidencebased.education/resource/retrieval-and-spaced-practice-study-strategies-that-must-be-combined/>

⁴ ⁵ Six Strategies for Effective Learning: A Summary for Teachers — The Learning Scientists

<https://www.learningscientists.org/blog/2019/11/28-1>

⁶ Is Multitasking More Efficient? | Newswise

<https://www.newswise.com/articles/is-multitasking-more-efficient>